# Interactive Graph-Based Distillation Integrated Meta-Learning Network for Hyperspectral Image Incremental Classification

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***Abstract*—Hyperspectral image (HSI) incremental classification aims to break the limitation of the fixed-category HSI classification models and implement adaptive recognition of new categories through incremental learning. The inherent characteristic of spectral variability in HSI aggravates distribution drift between new and old classes, and exacerbates the forgetting of old class knowledge in the HSI incremental classification. In this article, we propose the interactive graph-based distillation integrated meta-learning network (IGDIMN) for HSI incremental classification, which achieves the efficient memory transfer of old-class knowledge between the initial training phase and the incremental training phase (ILP). Specifically, we present the dual-graph network with dynamic distillation (DGNet-DD), which refines old-class knowledge in the incremental phase based on the hierarchical dual-graph interaction strategy. In the DGNet-DD, the proposed interactive graph distillation (IGD) module performs hierarchical distillation on two generated graphs that reflect instance-level and distribution-level relations of spatial–spectral features. The IGD module facilitates knowledge transfer between the initial training phase (ITP) and ILP stages, preserving learned knowledge and enhancing the capability to recognize new classes. Without relying on old-class exemplar replay, we further propose the prototype-aided memory transfer module (PAMTM) that leverages new-class prototypes to suppress distribution drift between old and new land-cover classes in the ILP. Besides, we adopt a multilevel prototype constraint to further enhance the knowledge transfer capability across different stages. Extensive experiments on three popular HSI datasets validate the superiority of the proposed IGDIMN method compared to other typical HSI incremental classification approaches.**

***Index Terms*—Graph neural networks (GNNs), hyperspectral image (HSI) incremental classification, knowledge distillation (KD), meta-learning (ML), prototype learning (PL).**

## I. Introduction

HYPERSPECTRAL image (HSI) captures fine-grained spectral features of ground objects through contiguous narrow bands, generating the spatially and spectrally integrated 3-D data cube with abundant spatial–spectral information [1]. As the core task in remote sensing interpretation, HSI classification [2], [3], [4] aims to predict semantic labels for individual pixels, which has been widely applied in the fields of agriculture investigation [5], [6], environmental monitoring [7], [8], marine detection [9], [10], and geological exploration [11], [12]. Early HSI classification research primarily focused on spectral feature optimization [13], [14], and spatial-context modeling [15], [16], independently capturing image features from the spectral dimension and spatial dimension to improve classification accuracy. The development of deep learning has significantly prompted HSI classification performance due to the distinct feature extraction capabilities and advanced contextual modeling mechanisms. Popular deep learning models including convolutional neural networks (CNNs) [17], [18], [19], Transformer [20], [21], and graph neural networks (GNNs) [22], [23], [24] have advanced the development of HSI classification. Typically, Ma et al. [25] proposed the hybrid CNN model via redundancy-removed multiscale 3-D spatial convolution and grouped spectral convolution for HSI classification. Xu et al. [26] proposed the cross-attention fusion network bridging CNN and Transformer (CAF-Former) model by integrating the local feature extraction and the global dependencies. Qin et al. [27] proposed the GCN-based semisupervised framework that leveraged graph adjacency matrices and spatial features to address scarce labeled data in HSI classification. Yu et al. [28] proposed the graph-polarized fusion (GPF) network to integrate the GNN branch and CNN branch, incorporating residual connections and weighted fusion to enhance HSI classification performance. Liu et al. [29] proposed the scaled dot-product center attention (SDPCA) mechanism to achieve joint optimization of spectral–spatial features for the HSI classification. The multiarea target attention (MATA) method [30] presented an asymmetric partitioning paradigm and combined a class-weighted decision mechanism to address the multiscale spatial adaptation problem of HSI classification. The SegHSI framework [31] constructed a globally segmented framework for HSI by clustering attention along with a teacher–student training strategy.

With the improvement of observation capabilities on remote sensing platforms, new land-cover classes continuously emerge during the acquisition of HSI. In such scenarios, traditional fixed-category HSI classification models face challenges

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in recognizing new classes while preserving previous knowledge, resulting in catastrophic forgetting [32]. To address the challenges, incremental learning [33], [34], [35] has become an effective approach for continuously learning new classes while retaining discriminative ability for old classes. The core task of the HSI incremental classification is to incorporate incremental learning that preserves discriminative ability for previously learned classes while recognizing new classes, aiming to address the catastrophic forgetting problem encountered by traditional models in incremental scenarios. Generally, the process of the HSI incremental classification consists of the initial training phase and the incremental training phase (ILP). Specifically, the initial training phase primarily involves training the model based on the initial class set to learn the spatial–spectral feature representations of old classes. The ILP aims to empower the model to recognize new classes while maintaining recognition ability for old classes.

In recent years, the incremental learning methods [36], [37], [38], [39], [40], [41], [42] have emerged in computer vision, benefiting the HSI incremental classification from the advancements. Notably, knowledge distillation (KD) [43], [44], [45] has been widely adopted in incremental classification on natural images [46], [47], [48], where it helps alleviate catastrophic forgetting by guiding the student model to mimic the behavior of the teacher model. The learning without forgetting (LwF) [46] that was regarded as milestone work pioneered integrating KD into incremental learning and demonstrated the effectiveness of KD in preserving old-class knowledge. Wang et al. [37] designed the feature boosting and compression for the class-incremental learning (FOSTER) method that employed feature distillation to mitigate catastrophic forgetting. Zhu et al. [38] proposed the prototype augmentation and self-supervision (PASS) method based on class prototypes, which memorizes and augments the prototypes of old classes to maintain learned knowledge. For incremental classification of remote sensing images, Zhao et al. [39] presented a deep metric continual learning approach that enabled continual updating and memory preservation in HSI classification via spectral–spatial distillation and multilevel metric optimization. Xu et al. [40] proposed retaining old-class knowledge while learning new land-cover types through KD and optimizing spatial–spectral information via channel attention to enable full-class classification of HSI. The approach of few-shot class-incremental learning (FSCIL) [41] mitigated the overfitting in few-shot incremental learning scenarios by decoupling feature extractor and classifier parameters while integrating cosine similarity metrics. The distillation-constrained prototype representation network (DCPRN) [42] mitigated catastrophic forgetting in HSI classification via prototype representation and dual KD, enhancing old-class discrimination and adaptive learning.

Although existing incremental learning methods have made progress in mitigating catastrophic forgetting, the inherent spectral variability exacerbates the distribution drift between old and new classes, which impedes knowledge transfer in HSI incremental classification. In this article, we propose the interactive graph-based distillation integrated meta-learning network (IGDIMN) for HSI incremental classification. Specifically, the proposed framework comprises the initial training phase (ITP) and the ILP. The spatial–spectral dual-branch network (SSDBN) is responsible for extracting multilevel spatial–spectral joint features, and the dual-graph network with dynamic distillation (DGNet-DD) conducts feature refinement through interactions between dual graphs in the ITP. Without relying on old-class exemplar replay, the ILP model (ILPM) enhances knowledge consolidation via the interactive graph-based distillation (IGD). Besides, the proposed model further presents the prototype-aided memory transfer module (PAMTM) to prompt the cross-stage knowledge transfer through new class prototypes. Collaboratively, the combined operation of IGD and PAMTM mitigates catastrophic forgetting, enhancing new-class discriminability and stabilizing old-class knowledge retention for HSI incremental classification.

The main contributions of the work are summarized as follows.

1) We propose the IGDIMN for HSI incremental classification. The proposed model integrates dual-graph distillation and prototype alignment to effectively mitigate catastrophic forgetting and alleviate distribution drift caused by spectral variability. To our knowledge, it is the first attempt to incorporate dual-graph distillation within the meta-learning framework to address the transfer of old-class knowledge and mitigate forgetting in the recognition of new categories for HSI. In addition, our model presents the multiloss optimization strategy in the meta-learning framework, which jointly improves new-class discriminability and the stability of old-class knowledge retention in HSI incremental classification.
2) We present the IGD to promote the spatial–spectral information propagation between nodes through dual-graph interaction, thereby strengthening retention of high-confidence knowledge from old classes and improving robustness against spectral variability of HSI. Building upon the interaction process, the dual-graph distillation loss is defined in our model to transfer discriminative knowledge of hyperspectral old-class samples from the ITP to the ILP, enabling our model to gradually consolidate the representation of old land-cover classes in HSI incremental classification.
3) We propose the PAMTM in the incremental learning stage that serves as the bridge between the ITP and ILP, generating the prototype-assisted loss through the multilevel prototype constraint. Notably, without relying on old-class sample replay, PAMTM guides the alignment of new-class prototypes derived from two graphs through the prototype-assisted loss, effectively alleviating distribution drift between HSI land-cover categories.

The remainder of this article is organized as follows. Section II describes the architecture of the proposed IGDIMN in detail, Section III presents experimental results and analysis, and Section IV provides the conclusion.

## II. Proposed Approach

Fig. 1 illustrates the overall architecture of IGDIMN, which is driven by the meta-learning framework to ensure effec-

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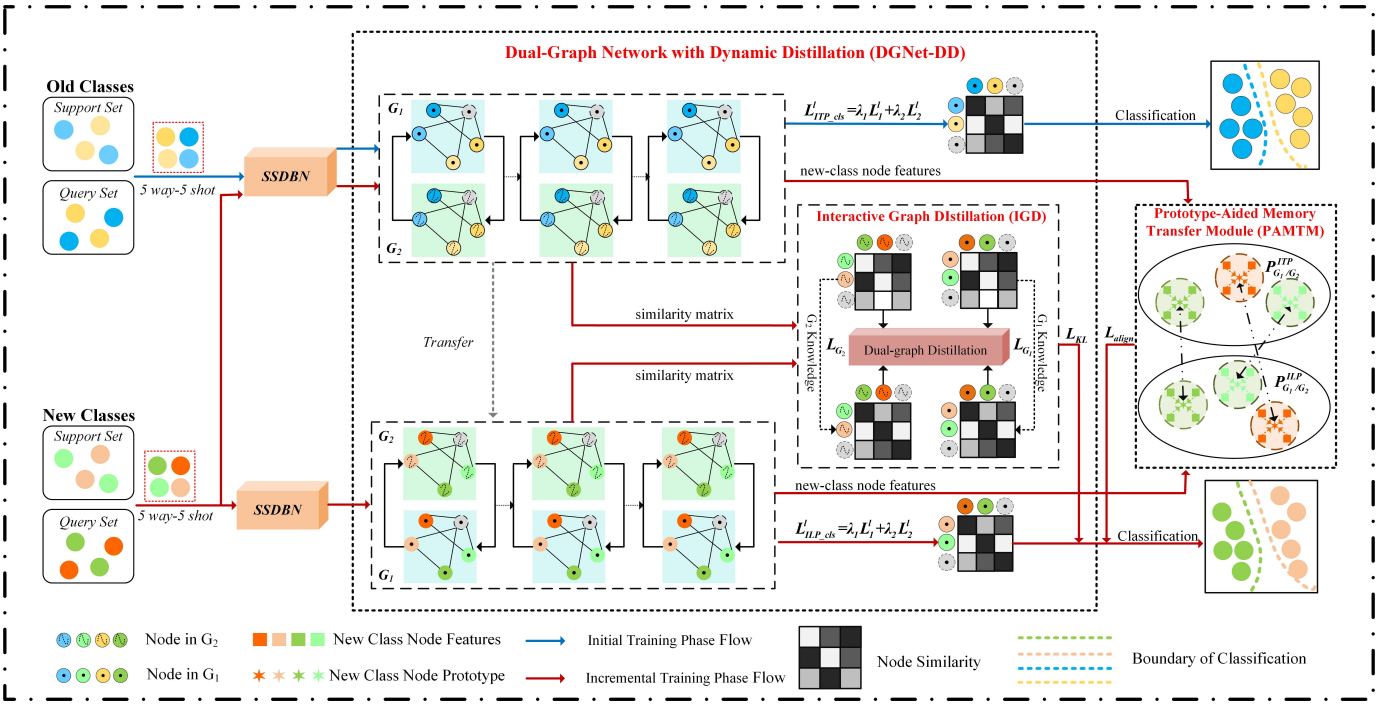



Fig. 1. Flowchart of the IGDIMN network with a two-phase framework. In the initial training phase, the initial training phase model is trained on selected old classes. In the ILP, the proposed ILP model enhances the representation of new-class features while alleviating catastrophic forgetting through the integration of dual-graph distillation and prototype alignment. The IGD enhances category-discriminative representations via dual-graph distillation, while the PAMTM facilitates cross-phase knowledge transfer between two phases.

tive adaptation to new classes. In the ITP, the ITP model (ITPM) captures discriminative spatial–spectral features of old-class samples through the SSDBN and constructs dual-graph structures via the DGNet-DD to preserve the optimal training parameters. In the ILP, the ILPM retains old-class knowledge through IGD without relying on old-class data replay. Meanwhile, the PAMTM collaborates with IGD to further enhance old-class knowledge transfer and effectively suppress distribution drift caused by the spectral variability of HSI data.

### A. Problem Definition

In the HSI incremental classification training process, the training dataset $D_{\text{train}}$ is divided into the old-class dataset $D_{\text{train}}^{\text{old}}$ and the new-class dataset $D_{\text{train}}^{\text{new}}$, where $D_{\text{train}} = D_{\text{train}}^{\text{old}} \cup D_{\text{train}}^{\text{new}}$. In the training phase, for the meta-learning framework, the support set $\mathcal{S} \subseteq D_{\text{train}}$ contains $N$ classes with $K$ samples per class (i.e., $N$-way-$K$-shot setting). In particular, $\mathcal{S}$ is defined as: $\mathcal{S} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_{N\times K}, y_{N\times K})\}$, where $x_i$ denotes the $i$th sample of the HSI, $y_i$ denotes the class label, $N$ denotes the number of classes, and $K$ denotes the number of samples per class. The query set $\mathcal{Q} \subseteq D_{\text{train}}$ is defined as: $\mathcal{Q} = \{(x_{N\times K+1}, y_{N\times K+1}), \ldots, (x_{N\times K+\widetilde{M}}, y_{N\times K+\widetilde{M}})\}$, where $\widetilde{M}$ represents the number of samples. Both $\mathcal{S}$ and $\mathcal{Q}$ provide labeled data in the training phase. In the testing phase, for the test dataset $D_{\text{test}}$, the support set $\mathcal{S} \subseteq D_{\text{test}}$ and the query set $\mathcal{S} \subseteq D_{\text{test}}$ satisfy $\mathcal{S} \cap \mathcal{Q} = \emptyset$. In this article, the incremental classification task aims to achieve classification over all categories through the two-stage learning process, in which the ITP phase trains the ITPM on $D_{\text{train}}^{\text{old}}$ to acquire knowledge of old classes, and the ILP phase trains the ILPM on $D_{\text{train}}^{\text{new}}$ to recognize new classes while retaining old-class knowledge in HSI incremental classification.

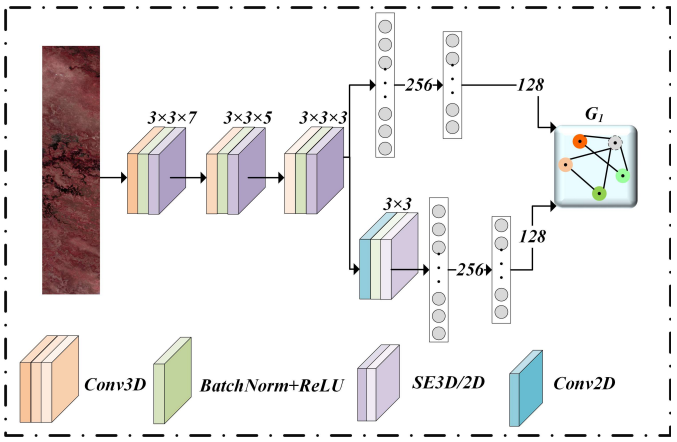



Fig. 2. Overview of the SSDBN architecture.

### B. Spatial–Spectral Dual-Branch Network

To effectively integrate the spectral and spatial features of HSI, we propose the SSDBN based on multiscale convolution and channel attention mechanisms. The detailed structure is summarized in Table I, and the overall architecture is illustrated in Fig. 2. The SSDBN performs spectral and spatial features learning through dual-path convolutional processing. For spectral feature extraction, the 3-D convolutional structure progressively employs three multistages to hierarchically capture spectral contextual information. By cascading convolutional

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TABLE I
BACKBONE STRUCTURE OF SSDBN

| | Layer | Input Size | Kernel Sets | Output Size |
|---|---|---|---|---|
| Feature Extractor | Conv3D+BN+ ReLU+SE3D | $10 \times 1$ $\times 20 \times 15 \times 15$ | $3 \times 3 \times 7$ $3 \times 3 \times 5$ $3 \times 3 \times 3$ | $10 \times 32$ $\times 8 \times 9 \times 9$ |
| | Conv2D+BN+ ReLU+SE2D | $10 \times 32$ $\times 8 \times 9 \times 9$ | $3 \times 3$ | $10 \times 64$ $\times 4 \times 4$ |
| | SE3DBlock | $10 \times 8$ $\times 14 \times 13 \times 13$ | compression ratio 16 (3D) | $10 \times 8$ $\times 14 \times 13 \times 13$ |
| | SE2DBlock | $10 \times 64$ $\times 4 \times 4$ | compression ratio 8 (2D) | $10 \times 64$ $\times 4 \times 4$ |

layers, the network achieves multiscale modeling and hierarchical representation of spectral features. Each level incorporates a 3-D squeeze-and-excitation channel attention module (SE3D) to dynamically adjust spectral channel weights. For spatial feature extraction, the 3-D spectral tensor is initially transformed into the 2-D spatial representation, where 2-D convolutional layers with 2-D squeeze-and-excitation channel attention module (SE2D) subsequently perform spatial downsampling to enhance discriminative local spatial patterns. The SSDBN simultaneously outputs normalized spectral and spatial features, which are fed into subsequent GNN modules.

### C. Dual-Graph Network With Dynamic Distillation

Motivated by [45] and [49], we present dual-graph interaction and KD approach, which constructed similarity matrices between two graphs ($G_1^l$ and $G_2^l$) to achieve robust knowledge retention for the HSI incremental classification.

*1) Dual-Graph Generation Process:* In our model, the DGNet-DD is composed of three-layer graph structures. The graph structure at the $l$th layer is realized through the interaction between the built graphs $G_1^l = (V_1^l, E_1^l)$ and $G_2^l = (V_2^l, E_2^l)$, where $V_1^l = \{v_1^{l,i}\}$, $E_1^l = \{e_1^{l,ij}\}$, $V_2^l = \{v_2^{l,i}\}$, and $E_2^l = \{e_2^{l,ij}\}$, with $i, j = 1, 2, \ldots, Z$ and $l \in \mathcal{L}$ representing the network layer index (e.g., $l = 1, 2, 3$), where $Z$ is the total number of nodes in the graph. $G_1^l$ is designed to capture instance-level relations between nodes, while $G_2^l$ is constructed to capture distribution-level relations between nodes, where the two graphs are iteratively updated with each other. Within each individual graph, the edge features of $G_1^l$ and $G_2^l$ are derived from the corresponding node features through an encoding network. The encoding network consists of two blocks, which are composed of Conv2d, Batch Normalization, and ReLU operations.

For $G_1^l$, the initial node feature $v_1^{l,0}$ is obtained from the SSDBN, and the edge feature $e_1^{l,ij}$ is computed based on the similarity between node features, as defined in the following equation:

$$e_1^{l,ij} = \begin{cases} F_1^l\left(\left(v_1^{l-1,i} - v_1^{l-1,j}\right)^2\right) \cdot e_1^{l-1,ij}, & \text{if } l > 0 \\ F_1^0\left(\left(v_1^{0,i} - v_1^{0,j}\right)^2\right), & \text{if } l = 0 \end{cases} \tag{1}$$

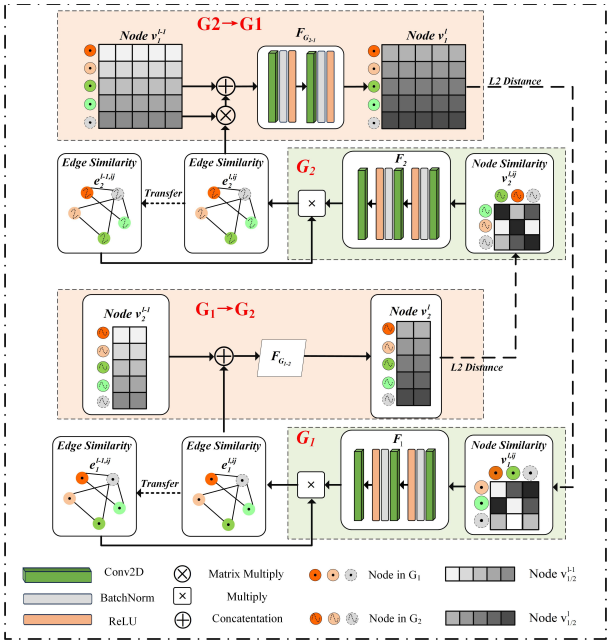



Fig. 3. Cross-graph aggregation process between $G_1$ and $G_2$.

where $F_1^l$ refers to the aforementioned encoding network composed of two blocks, with the Conv2d, Batch Normalization, and ReLU operations.

For $G_2^l$, the edge feature $e_2^{l,ij}$ is defined as shown in the following equation:

$$e_2^{l,ij} = \begin{cases} F_2^l\left(\left(v_2^{l-1,i} - v_2^{l-1,j}\right)^2\right) \cdot e_2^{l-1,ij}, & \text{if } l > 0 \\ F_2^0\left(\left(v_2^{0,i} - v_2^{0,j}\right)^2\right), & \text{if } l = 0. \end{cases} \tag{2}$$

*2) Dual-Graph Update Strategy:* $G_1^l$ and $G_2^l$ are not two independently learning networks with heterogeneous knowledge. Actually, the generation of $G_1^l$ and $G_2^l$ is an alternating iterative process. During the interaction, fine-grained instance similarity captured by $G_1^l$ is aggregated and employed to update node features of $G_2^l$, thereby elevating local information to global category distribution representation. In turn, the distribution relationships between categories obtained by $G_2^l$ are fed back to $G_1^l$ for node feature updating.

In the interaction between the two graphs, the edge feature $E_1^l$ of $G_1^l$ participates in deriving the node feature $V_2^l$ of $G_2^l$, and the aggregation from $G_1^l$ to $G_2^l$ is achieved by the aggregation network $F_{G_{1-2}}$, which is specifically composed of the fully connected layer and the nonlinear activation function ReLU. The edge feature $E_2^l$ of $G_2^l$ participates in deriving the node feature $V_1^l$ of $G_1^l$, and the aggregation network $F_{G_{2-1}}$ from $G_2^l$ to $G_1^l$ is composed of two blocks with the Conv2d, Batch Normalization, and ReLU layers. The interaction process of the two graphs within the three-layer graph structure is summarized as: $E_1^l \rightarrow V_2^l \rightarrow E_2^l \rightarrow V_1^l \rightarrow E_1^{l+1}$, with the cross-graph aggregation process between $G_1^l$ and $G_2^l$ detailed in Fig. 3.

For the aggregation from $G_1^l$ to $G_2^l$, distribution-level information among samples is captured by integrating instance-

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level similarity from $G_1^l$. Node features in $G_2^l$ are constructed through edge features $e_1^{l,ij}$ from $G_1^l$ combined with initial node features $v_2^{0,i}$ from $G_2^l$, as defined in the following equations:

$$v_2^{0,i} = \begin{cases} \Big\|_{j=1}^{M} \delta\left(y_i, y_j\right), & \text{if } x_i \text{is labeled} \\ \left[\frac{1}{M}, \ldots, \frac{1}{M}\right], & \text{otherwise} \end{cases} \tag{3}$$

$$v_2^{l,i} = F_{G_{1--2}}\left(\Big\|_{j=1}^{M} e_1^{l,ij}, v_2^{l-1,i}\right) \tag{4}$$

where $M$ denotes the number of labeled samples in the classification task; $\|$ represents the concatenation operation; and $\delta(\cdot)$ is the Kronecker delta function, which returns 1 if $y_i = y_j$, and 0 otherwise.

For the aggregation from $G_2^l$ to $G_1^l$, to further optimize node features $v_1^{l,i}$ in $G_1^l$, edge features $e_2^{l,ij}$ of $G_2^l$ are employed to aggregate neighboring node features for updating

$$v_1^{l,i} = F_{G_{2-1}}\left(\sum_{j=1}^{T}\left(e_2^{l,ij} \cdot v_1^{l-1,j}\right), v_1^{l-1,i}\right). \tag{5}$$

Through (5), node features integrate distribution information to proceed with next layer updates. Overall, the algorithm of DGNet-DD is briefly outlined in Algorithm 1.

**Algorithm 1** Algorithm of DGNet-DD

**Input:** Initialize $v_1^{l,0}$, $e_1^{0,ij}$ with SSDBN, L, Z
**Output:** $E_{G_1}^l, E_{G_2}^l, P_{G_1}^{ILP}, P_{G_2}^{ILP}$

```
Initialize:
v_2^{0,i} ← Calculate G_2^l Node features with Equ. (3)
e_2^{0,ij} ← Calculate G_2^l Edge features with Equ. (2)
For l = 1 to L do
    For i = 1 to Z do
        e_1^{l,ij} ← Calculate G_1^l Edge features with Equ. (1)
    End For
    For i = 1 to Z do
        v_2^{l,i} ← Calculate G_2^l Node features with Equ. (4)
    End For
    For i = 1 to Z do
        e_2^{l,ij} ← Calculate G_2^l Edge features with Equ. (2)
    End For
    For i = 1 to Z do
        v_1^{l,i} ← Calculate G_1^l Node features with Equ. (5)
    End For
End For
```

*3) Interactive Graph-Based Distillation:* To enhance knowledge retention in incremental learning, the IGD is proposed to guarantee the effective preservation of previously learned class knowledge without relying on exemplar replay in the ILP. Notably, the IGD constructs the cross-phase KD based on the dual-graph update strategy, leveraging similarity matrices obtained from both the ITPM and the ILPM.

As shown in Fig. 4, IGD leverages the ITPM and the ILPM to first generate similarity matrices $E_{G_1,\text{tea}}^l, E_{G_1,\text{stu}}^l$ for $G_1$, and $E_{G_2,\text{tea}}^l, E_{G_2,\text{stu}}^l$ for $G_2$. Next, similarity matrices $E_{G_1,\text{tea}}^l$ and $E_{G_2,\text{tea}}^l$ estimate confidence scores to guide the adaptive adjustment of the distillation temperature. The relatively low confidence score leads to the higher temperature to smooth the feature distribution, whereas the high-confidence score results in the lower temperature to emphasize the retention of knowledge, thereby enhancing the ability of ILPM to acquire transferable knowledge. The distillation temperature $T_{G,i}$ is calculated as shown in the following equation:

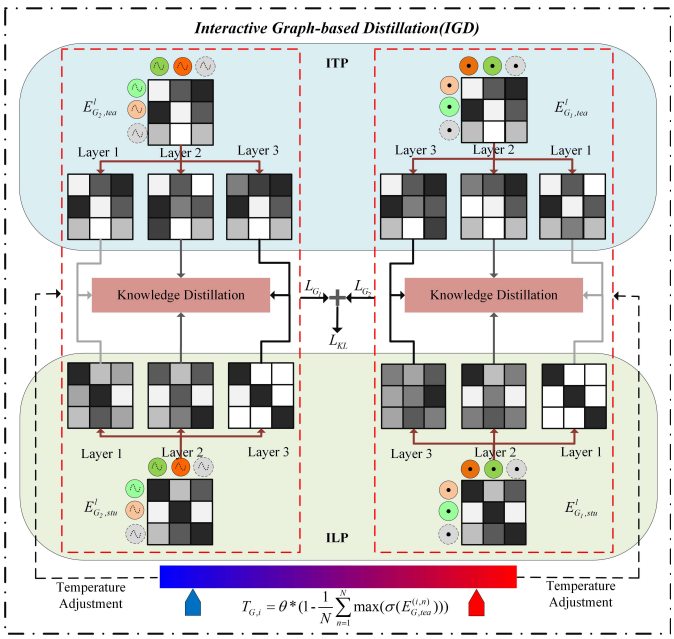



Fig. 4. Diagram of the interactive graph-based distillation module.

$$T_{G,i} = \theta \cdot \left(1 - \frac{1}{N}\sum_{n=1}^{N} \max\left(\sigma\left(E_{G,\text{tea}}^{(i,n)}\right)\right)\right) \tag{6}$$

where $G = \{G_1, G_2\}$, $\theta$ is empirically set to 3, the function $\max(\cdot)$ returns the confidence score of the node as the highest probability, and $\sigma(\cdot)$ denotes the SoftMax operation.

In addition, considering the three-layer graph structure of DGNet-DD, the hierarchical distillation strategy is applied to similarity matrices generated at different layers. The similarity matrices from the first two layers, which contain more redundant information, are assigned lower weights, while the matrix from the last layer is given a higher weight, leading to the effective distillation outcome. The calculation of $L_{\text{KL}}$ is shown in the following equations:

$$\delta_{\text{KL}}^{(i,n,k)} = \log\sigma\left(\frac{E_{G,\text{tea}}^{(i,n,k)}}{T_{G,i}}\right) - \log\sigma_{\log}\left(\frac{E_{G,\text{stu}}^{(i,n,k)}}{T_{G,i}}\right) \tag{7}$$

$$D_{\text{KL}}^{(G,i)} = \frac{1}{N^2}\sum_{n=1}^{N}\sum_{k=1}^{N}\sigma\left(\frac{E_{G,\text{tea}}^{(i,n,k)}}{T_{G,i}}\right) \cdot \delta_{\text{KL}}^{(i,n,k)} \tag{8}$$

$$L_G = \sum_{i=1}^{l} \lambda_i \cdot D_{\text{KL}}^{(G,i)} \cdot T_{G,i}^2, \quad \lambda_i = \begin{cases} 0.1, & \text{if } i < L \\ 1, & \text{if } i = L \end{cases} \tag{9}$$

$$L_{\text{KL}} = \alpha L_{G_1} + \beta L_{G_2} \tag{10}$$

where $\sigma_{\log}(\cdot)$ represents the log-SoftMax operation; $T_{G,i}$ denotes the distillation temperature; $l$ is the number of layers in

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the DGNet-DD; and $\alpha$ and $\beta$ denote the weighting coefficients for the loss corresponding to $G_1$ and $G_2$, respectively.

### D. Prototype-Aided Memory Transfer Module

We present the PAMTM to further strengthen old-class knowledge transfer and mitigate distribution shift. In the ILP, new class samples are separately fed into the ITPM and the ILPM to extract class prototypes from both phases. Specifically, the prototypes $P_{G_1}^{\mathrm{ITP}}$ and $P_{G_2}^{\mathrm{ITP}}$ are obtained from ITPM, and $P_{G_1}^{\mathrm{ILP}}$ and $P_{G_2}^{\mathrm{ILP}}$ are generated by ILPM. The extracted prototypes from both phases are normalized independently as shown in the following equation:

$$P_{G_1/G_2}^{\mathrm{ITP}} = \frac{P_{G_1/G_2}^{\mathrm{ITP}}}{\left\| P_{G_1/G_2}^{\mathrm{ITP}} \right\|_2}, \quad P_{G_1/G_2}^{\mathrm{ILP}} = \frac{P_{G_1/G_2}^{\mathrm{ILP}}}{\left\| P_{G_1/G_2}^{\mathrm{ILP}} \right\|_2}. \tag{11}$$

Furthermore, PAMTM leverages the multilevel prototype alignment mechanism to perform cross-stage alignment between prototypes from $G_1^l$ and $G_2^l$. To enhance alignment stability and improve the efficiency of knowledge transfer, different weights are assigned to the prototype alignment losses of each graph. Without relying on old-class sample replay, PAMTM facilitates continuous migration and representation of old-class knowledge between the ITP and ILP. The $L_G^{\mathrm{align}}$ for a graph $G$ is calculated as shown in (12). The prototype alignment loss $L_{\mathrm{align}}$ combines the losses from $G_1^l$ and $G_2^l$, as defined in the following equation:

$$L_G^{\mathrm{align}} = 1 - \frac{1}{N} \sum_{i=1}^{N} \cos\left(P_G^{\mathrm{ITP},i}, P_G^{\mathrm{ILP},i}\right) \tag{12}$$

$$L_{\mathrm{align}} = L_{G_1}^{\mathrm{align}} + \lambda_{G_2} L_{G_2}^{\mathrm{align}} \tag{13}$$

where $\lambda_{G_2}$ denotes the weighting coefficient for loss corresponding to $G_2$, which is empirically fixed at 0.1 in this article.

### E. Classification Process

In the ITP, our model executes feature extraction on old-class samples from support set and query set within dataset $D_{\mathrm{train}}^{\mathrm{old}}$. Each query node in the graph is classified based on the final-layer outputs of DGNet-DD via the cross-entropy loss. The classification losses of $G_1$ and $G_2$, denoted as $L_1^l$ and $L_2^l$ based on the cross-entropy loss, are defined in the following equations:

$$L_1^l = L_{\mathrm{CE}}\left(\mathrm{Softmax}\left(\sum_{j=1}^{M} e_1^{l,ij}, \mathrm{one_hot}\left(y_j\right)\right), y_i\right) \tag{14}$$

$$L_2^l = L_{\mathrm{CE}}\left(\mathrm{Softmax}\left(\sum_{j=1}^{M} e_2^{l,ij}, \mathrm{one_hot}\left(y_j\right)\right), y_i\right) \tag{15}$$

where $L_{\mathrm{CE}}$ denotes the cross-entropy loss, $y_j$ represents the label of the $j$th sample in the support set, $y_i$ indicates the true label of the query set, $e_1^{l,ij}$ corresponds to the edge features of the $l$th layer in $G_1$, and $e_2^{l,ij}$ defines the edge features of $G_2$ at the $l$th layer.

**Algorithm 2** Algorithm of IGDIMN

Initial Training Phase (ITP)
**Input:** $D_{train}^{old}$, N way-K shot
**Output:** $\theta_{ITP}, \phi_{ITP}$
**For** $n$ = 1 to Epoch **do**
**For** $b$ = 1 to Batch **do**
$L_{cls}^l \leftarrow$ Calculate the loss with Equ. (15)
$\theta, \phi =\leftarrow -\nabla_{\theta,\phi}(L_{cls}^l)$
**End For**
$\theta_{ITP}, \phi_{ITP} =\leftarrow$ Parameters of $SSDBN_\theta$ and $DGNet - DD_\phi$
**End For**
Incremental Training Phase (ILP)
**Input:** $D_{train}^{new}$, N way-K shot
**Output:** $\theta_{ILP}, \phi_{ILP}$
**For** $n$ = 1 to Epoch **do**
**For** $b$ = 1 to Batch **do**
$L_{cls'}^l \leftarrow$ Calculate the loss with Equ. (16)
$L_{KL} \leftarrow$ Calculate the loss with Equ. (10)
$L_{align} \leftarrow$ Calculate the loss with Equ. (13)
$L \leftarrow$ Calculate the loss with Equ. (17)
$\theta, \phi =\leftarrow -\nabla_{\theta,\phi}(L_{cls'}^l)$
**End For**
$\theta_{ILP}, \phi_{ILP} =\leftarrow$ Parameters of $SSDBN_\theta$ and $DGNet - DD_\phi$
**End For**

The overall ITP classification loss $L_{\mathrm{ITP_cls}}^l$ is defined in the following equation:

$$L_{\mathrm{ITP_cls}}^l = \lambda_1 L_1^l + \lambda_2 L_2^l \tag{16}$$

where $l$ denotes the total number of layers in the DGNet-DD and $\lambda_1$ and $\lambda_2$ represent the weight coefficients.

In the ILP, new-class samples $D_{\mathrm{train}}^{\mathrm{new}}$ are fed into both the ILPM and the ITPM, without relying on old-class sample replay. The IGD guides IGDIMN to focus on high-confidence regions by constructing alignment constraints between similarity matrices from different stages. Meanwhile, PAMTM achieves prototype alignment with the aid of the prototype alignment loss. Ultimately, the ILPM jointly optimizes the graph distillation loss and the prototype transfer loss in addition to the classification loss, thereby maximizing the retention of old-class knowledge and improving the discriminative ability for new classes.

The overall loss function of the ILP is defined in the following equation:

$$L = \sum_{l=1}^{\hat{l}} L_{\mathrm{ILP_cls}}^l + L_{\mathrm{KL}} + L_{\mathrm{align}} \tag{17}$$

where $\hat{l}$ denotes the total number of layers in the dual-graph network.

In the testing stage, we utilize the trained ILPM to classify all land-cover categories in HSI, including both old and new classes. The ILPM integrates the spectral–spatial knowledge acquired from previous learning phases, ensuring effective generalization to new land-cover types while mitigating the

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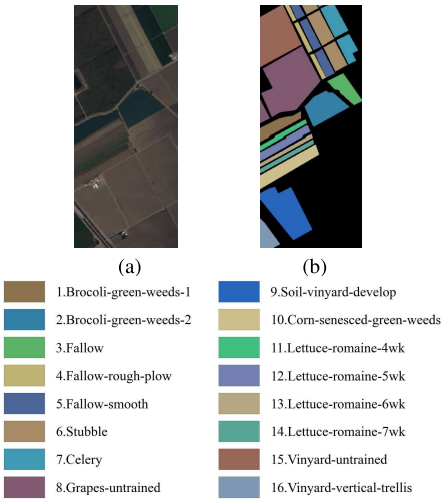



Fig. 5. Data of Salinas. (a) False-color image. (b) Ground-truth image.

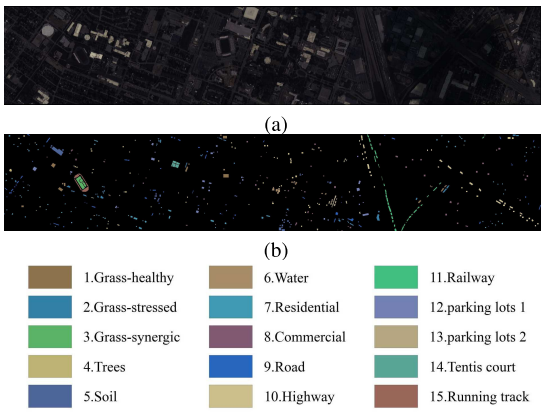



Fig. 6. Data of Houston. (a) False-color image. (b) Ground-truth image.

forgetting of previously seen classes. Furthermore, our model alleviates the distribution drift caused by spectral variability across different land-cover regions.

## III. Experiment and Analysis

### A. Data Description

This article employs three popular HSI datasets to validate the effectiveness of the proposed method.

*1) Salinas:* The Salinas dataset was captured by the AVIRIS sensor in Salinas Valley, California. The dataset has the spatial resolution of 3.7 m and the size of $512 \times 217$ pixels. The original data consists of 224 bands, and after removing 20 bands with severe water absorption, 204 bands remain. The study area includes 16 types of land cover classes. The false-color and ground-truth images are illustrated in Fig. 5(a) and (b), respectively.

*2) Houston:* The Houston dataset was collected by the ITRES CASI-1500 sensor over Houston, Texas, USA, and surrounding rural areas, covering the University of Houston campus. The dataset has a size of $349 \times 1905$ pixels with 144 bands. The study area contains 15 land-cover classes, with false-color and ground-truth images, respectively, shown in Fig. 6(a) and (b).

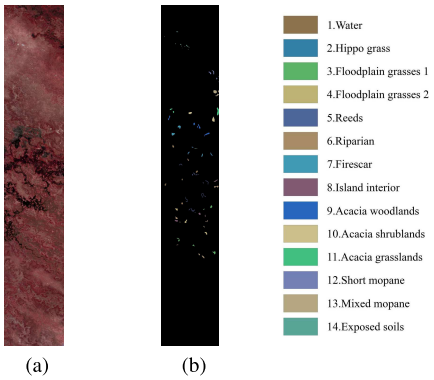



Fig. 7. Data of Botswana. (a) False-color image. (b) Ground-truth image.

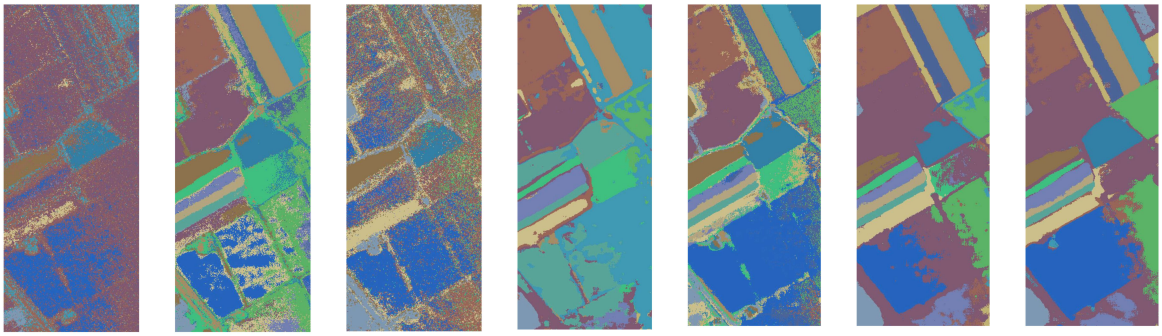
*3) Botswana:* The Botswana dataset was captured over the Okavango Delta, Botswana, by the Hyperion sensor on NASA's EO-1 satellite between 2001 and 2004. Comprising $256 \times 1476$ pixels, the dataset includes 145 spectral bands spanning 400–2500 nm, enabling detailed analysis of 14 land-cover classes in seasonal swamps, occasional swamps, and drier woodlands. The false-color and ground-truth images of Botswana are illustrated in Fig. 7(a) and (b), respectively.

### B. Experimental Configuration

All experiments ran on PyTorch 2.4 and Python 3.12 across systems featuring the Nvidia GTX 4060 GPU, Intel i9-14900HX processor, and 16-GB RAM. During task-driven meta-optimization, the dataset is initially divided into $15 \times 15$ patches, followed by PCA for preprocessing, where the top 20 principal components are retained. To enhance model robustness against hyperspectral perturbations, the self-supervised augmentation strategy is applied in the preprocessing stage. Sample diversity is increased across spatial and spectral dimensions. Spatial augmentation includes image rotation {90°, 180°, 270°} and random flipping, while spectral augmentation involves random channel dropout and injection of Gaussian noise, which strengthens the model against external interference. For each of the three datasets, 10% of the samples are selected as the training set, and the entire dataset serves as the test set for evaluation and result presentation. A 5way-5shot training setting is adopted in the meta-learning framework.

In addition, according to the convergence speed, the ITP for the Houston, Botswana, and Salinas datasets is conducted with 200 epochs and a learning rate of 0.001. The ILP is set to 1000 epochs with a learning rate of 0.0001 for all datasets. Adam optimizer is adopted to guide the parameter update optimization. In our experiment, overall accuracy (OA) and Kappa served as objective criteria for evaluating the performance of the method. To further measure the effectiveness of the experiment, OA was divided into four specific objective criteria [42] for different categories across different stages, including OA-INIT, OA-INIL, OA-ICIL, and OA-ILSM. To clearly assess the forgetting problem of old classes, we define the average forgetting rate (AFR), which is calculated by subtracting OA-INIL from OA-INIT. To demonstrate the classification performance of the proposed IGDIMN network, IGDIMN is compared with the series of other state-of-the-art

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(a) (b) (c) (d) (e) (f) (g)

Fig. 8. Classification maps obtained via different HSI incremental classification methods for the Salinas data. (a) LwF. (b) iCaRL. (c) End2End. (d) PASS. (e) FOSTER. (f) DCPRN. (g) IGDIMN.

TABLE II
COMPARISON OF CLASS PERFORMANCE OF DIFFERENT APPROACHES FOR THE SALINAS DATA

| | Class | LwF | iCaRL | End2End | PASS | FOSTER | DCPRN | IGDIMN |
|---|---|---|---|---|---|---|---|---|
| Initial Classes | Broccoli-green-weeds-1 | 92.43 ± 2.87 | 97.95 ± 3.53 | 93.50 ± 2.01 | 0.56 ± 1.26 | 99.25 ± 0.36 | 97.00 ± 1.28 | **99.90 ± 0.00** |
| | Broccoli-green-weeds-2 | 87.95 ± 6.34 | 97.70 ± 2.35 | 18.03 ± 28.49 | 12.41 ± 25.43 | 93.83 ± 6.23 | **98.17 ± 0.74** | 98.16 ± 0.62 |
| | Fallow-rough-plow | 20.83 ± 17.85 | 66.95 ± 13.93 | 33.76 ± 36.70 | 25.77 ± 34.96 | 77.80 ± 9.88 | 96.77 ± 0.39 | **99.25 ± 0.64** |
| | Soil-vineyard-develop | 74.44 ± 10.68 | 84.57 ± 10.82 | 77.67 ± 36.70 | 5.45 ± 5.06 | 92.17 ± 4.28 | 1.56 ± 0.44 | **92.92 ± 0.25** |
| | Corn-senesced-green-weeds | 45.63 ± 12.30 | 81.49 ± 8.83 | 11.41 ± 2.80 | 59.21 ± 18.45 | 51.86 ± 29.35 | 32.32 ± 0.69 | **91.10 ± 4.57** |
| | Lettuce-romaine-5wk | 6.42 ± 3.71 | 67.65 ± 16.33 | 0.00 ± 0.00 | 7.40 ± 15.33 | 68.45 ± 6.20 | 52.15 ± 0.28 | **96.08 ± 2.45** |
| | Vinyard-vertical-trellis | 41.82 ± 26.68 | 55.87 ± 16.34 | 25.78 ± 2.46 | 77.43 ± 13.42 | 71.66 ± 22.38 | 62.93 ± 11.91 | **98.53 ± 1.22** |
| | Fallow | 10.13 ± 12.27 | 57.98 ± 24.83 | 2.91 ± 2.66 | 4.46 ± 6.09 | 43.59 ± 13.36 | 68.70 ± 4.68 | **98.55 ± 1.15** |
| | Fallow-smooth | 25.37 ± 14.81 | 50.14 ± 22.77 | 4.56 ± 4.05 | 3.76 ± 4.46 | 66.61 ± 12.44 | 91.02 ± 5.81 | **98.33 ± 0.46** |
| Incremental Classes | Stubble | 11.98 ± 12.58 | 99.75 ± 0.40 | 82.98 ± 1.93 | 99.88 ± 0.16 | 99.93 ± 0.03 | **99.97 ± 0.00** | 96.29 ± 1.90 |
| | Celery | 13.51 ± 10.40 | 94.47 ± 4.78 | 51.67 ± 36.27 | 99.32 ± 0.85 | 99.53 ± 0.39 | **99.93 ± 0.02** | 98.54 ± 0.05 |
| | Grapes-untrained | 74.91 ± 8.53 | 83.79 ± 1.85 | 32.12 ± 6.08 | 92.19 ± 2.15 | 81.83 ± 8.86 | **96.67 ± 0.52** | 70.92 ± 1.48 |
| | Lettuce-romaine-4wk | 1.27 ± 2.61 | 99.08 ± 0.79 | 1.18 ± 2.64 | 98.15 ± 1.88 | 97.90 ± 1.54 | 99.72 ± 0.00 | **99.83 ± 0.18** |
| | Lettuce-romaine-6wk | 0.24 ± 0.42 | 98.72 ± 0.85 | 1.16 ± 0.69 | 99.30 ± 0.81 | 99.27 ± 0.23 | 94.83 ± 5.74 | **99.82 ± 0.10** |
| | Lettuce-romaine-7wk | 0.08 ± 0.19 | 97.74 ± 0.73 | 0.65 ± 0.29 | 98.21 ± 1.09 | 98.05 ± 0.91 | 92.86 ± 6.71 | **99.65 ± 0.39** |
| | Vinyard-untrained | 16.63 ± 9.76 | 85.70 ± 3.34 | 5.40 ± 3.16 | **88.06 ± 2.30** | 80.96 ± 9.97 | 80.34 ± 2.55 | 83.69 ± 1.33 |
| Performance | OA-INIT | 84.65 ± 5.59 | 86.98 ± 3.37 | 96.90 ± 0.07 | **99.95 ± 0.02** | 83.92 ± 5.35 | 99.70 ± 0.04 | 78.75 ± 0.03 |
| | OA-INIL | 53.20 ± 4.81 | 77.07 ± 5.23 | 35.45 ± 6.27 | 19.32 ± 5.33 | 77.03 ± 3.73 | 56.22 ± 1.07 | **95.51 ± 0.99** |
| | OA | 44.20 ± 2.10 | 83.66 ± 2.66 | 33.32 ± 5.53 | 59.38 ± 2.22 | 82.94 ± 1.45 | 76.22 ± 0.03 | **89.55 ± 0.40** |
| | AFR | 31.45 ↓ | 9.91 ↓ | 61.45 ↓ | 80.63 ↓ | 6.89 ↓ | 43.48 ↓ | **16.76** ↑ |
| | Kappa-ITP | 81.97 ± 6.61 | 84.89 ± 3.86 | 96.40 ± 0.08 | **99.95 ± 0.05** | 81.26 ± 6.22 | 99.65 ± 0.05 | 99.77 ± 0.02 |
| | Kappa-ILP | 34.43 ± 2.34 | 81.90 ± 2.92 | 26.17 ± 5.56 | 53.59 ± 2.97 | 81.12 ± 1.52 | 73.71 ± 0.02 | **96.87 ± 0.48** |

models on the three datasets, including FOSTER [37], PASS [38], DCPRN [42], LwF [46], iCaRL [50], and End2End [51].

### C. Comparative Results and Analysis

In the experiment on the Salinas dataset, nine classes are selected in the initial stage, namely, Brocoli-green-weeds-1, Brocoli-green-weeds-2, Fallow-rough-plow, Soil-vinyard-develop, Corn-senesced-green-weeds, Lettuce-romaine-5wk, Vinyard-vertical-trellis, Fallow, and Fallow-smooth. The classification results are summarized in Table II, while the colorful result maps for all models are illustrated in Fig. 8 to provide the visual comparison. The classification accuracy of IGDIMN in 11 classes outperforms that of other models. The accuracy rates of the initial classes all exceed 90%, indicating that the method effectively alleviates the problem of forgetting the information of the initial classes. Among the comparative models, the overall classification results of LwF and End2End are not entirely satisfactory. Similarly, the classification maps and accuracy generated bypass are also not satisfactory. IGDIMN achieves the OA of 89.55% and the Kappa coefficient of 96.87%, both showing clear improvements over DCPRN. Notably, the proposed IGDIMN achieves improved accuracy for the old classes and produces a positive AFR, which indicates effective retention of known knowledge during incremental learning in the Salinas data. Although iCaRL and FOSTER have relatively high comprehensive performance, the memory ability for the initial classes is limited. The experimental results show that by combining IGD and PAMTM, IGDIMN effectively balances the learning conflicts between new and old classes, thus alleviating the forgetting of knowledge of old classes while maintaining the adaptability to new classes.

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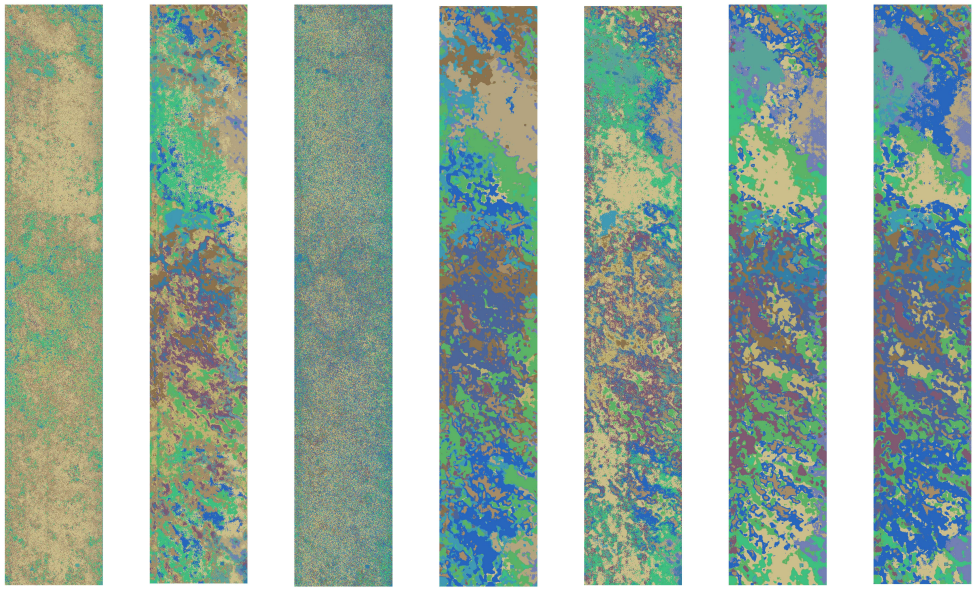



Fig. 9. Classification maps obtained via different HSI incremental classification methods for the Botswana data. (a) LwF. (b) iCaRL. (c) End2End. (d) PASS. (e) FOSTER. (f) DCPRN. (g) IGDIMN.

TABLE III
COMPARISON OF CLASS PERFORMANCE OF DIFFERENT APPROACHES FOR THE BOTSWANA DATA

| | Class | LwF | iCaRL | End2End | PASS | FOSTER | DCPRN | IGDIMN |
|---|---|---|---|---|---|---|---|---|
| Initial Classes | Acacia grasslands | 20.58 ± 12.42 | 75.20 ± 13.81 | 27.78 ± 12.22 | 5.02 ± 6.24 | 60.29 ± 17.85 | **99.35 ± 0.40** | 94.03 ± 4.41 |
| | Island interior | 14.63 ± 11.13 | 55.08 ± 20.02 | 14.97 ± 13.83 | 10.82 ± 12.09 | 49.07 ± 17.94 | **99.34 ± 0.24** | 87.88 ± 13.47 |
| | Mixed mopane | 17.43 ± 11.27 | 81.99 ± 11.97 | 29.96 ± 13.02 | 94.27 ± 6.62 | 89.88 ± 2.85 | 75.68 ± 26.98 | **98.21 ± 1.10** |
| | Exposed soils | 4.94 ± 4.88 | 61.41 ± 13.08 | 23.29 ± 18.06 | 67.53 ± 7.09 | 44.47 ± 14.73 | **99.82 ± 0.00** | 94.95 ± 4.79 |
| | Hippo grass | 12.31 ± 8.70 | 72.31 ± 4.62 | 10.33 ± 20.67 | 49.89 ± 7.15 | 87.47 ± 14.17 | **98.90 ± 0.00** | 98.81 ± 1.29 |
| | Riparian | 8.68 ± 4.83 | 66.86 ± 13.97 | 13.06 ± 5.34 | 71.07 ± 13.13 | 54.46 ± 14.45 | **88.84 ± 6.22** | 81.79 ± 11.42 |
| | Floodplain grasses 2 | 12.16 ± 9.48 | 83.71 ± 15.98 | 16.29 ± 3.80 | 60.21 ± 26.62 | 67.11 ± 10.34 | **99.48 ± 0.00** | 99.44 ± 1.01 |
| | Acacia shrublands | 14.35 ± 7.42 | 85.38 ± 12.60 | 37.22 ± 11.53 | 25.82 ± 35.40 | 74.77 ± 7.60 | **94.08 ± 8.30** | 92.58 ± 6.79 |
| Incremental Classes | Water | 16.30 ± 21.58 | **96.79 ± 2.99** | 7.49 ± 4.55 | 96.54 ± 3.06 | 96.38 ± 2.70 | 95.80 ± 3.58 | 96.22 ± 2.59 |
| | Acacia woodlands | 28.34 ± 4.91 | 77.17 ± 11.09 | 17.95 ± 15.61 | 86.57 ± 6.71 | 88.25 ± 10.18 | 77.39 ± 6.03 | **98.79 ± 1.30** |
| | Reeds | 22.23 ± 13.71 | 47.19 ± 13.68 | 21.07 ± 22.03 | **94.63 ± 4.52** | 38.84 ± 14.14 | 82.23 ± 7.19 | 87.36 ± 5.17 |
| | Short mopane | 8.10 ± 12.67 | 52.02 ± 16.71 | 2.21 ± 2.76 | 33.50 ± 22.56 | 84.05 ± 12.82 | 73.37 ± 21.14 | **95.80 ± 3.80** |
| | Floodplain grasses 1 | 6.46 ± 9.60 | 66.73 ± 11.48 | 2.21 ± 1.36 | 96.81 ± 1.81 | 80.62 ± 9.52 | 92.57 ± 4.02 | **99.20 ± 0.80** |
| | Firescar | 11.76 ± 11.54 | 89.27 ± 11.54 | 11.50 ± 6.66 | 95.54 ± 3.70 | **98.20 ± 1.37** | 95.97 ± 2.14 | 97.61 ± 3.92 |
| Performance | OA-INIT | 48.31 ± 4.03 | 80.50 ± 4.07 | 44.09 ± 2.74 | **100.00 ± 0.00** | 79.20 ± 5.13 | **100.00 ± 0.00** | **100.00 ± 0.00** |
| | OA-INIL | 13.68 ± 4.15 | 74.53 ± 4.90 | 22.97 ± 7.87 | 46.68 ± 7.97 | 68.74 ± 3.07 | **93.69 ± 5.62** | 92.83 ± 1.48 |
| | OA | 15.01 ± 2.00 | 73.84 ± 2.90 | 17.36 ± 3.48 | 65.68 ± 4.67 | 73.43 ± 0.88 | 90.48 ± 2.42 | **94.27 ± 0.63** |
| | AFR | 34.63 ↓ | **5.97** ↓ | 21.12 ↓ | 53.32 ↓ | 10.46 ↓ | 6.31 ↓ | 7.17 ↓ |
| | Kappa-ITP | 39.20 ± 4.71 | 77.45 ± 4.71 | 34.14 ± 3.24 | **100.00 ± 0.00** | 76.02 ± 5.86 | **100.00 ± 0.00** | **100.00 ± 0.00** |
| | Kappa-ILP | 7.37 ± 2.38 | 71.70 ± 3.13 | 9.96 ± 3.79 | 62.73 ± 5.10 | 71.31 ± 0.92 | 89.70 ± 2.62 | **93.80 ± 0.68** |

The second experiment is conducted on the Botswana dataset. In the ITP, eight classes are selected as the initial classes, namely, Acacia grasslands, Island interior, Mixed mopane, Exposed soils, Hippo grass, Riparian, Floodplain grasses 2, and Acacia shrublands. As shown in Table III and Fig. 9, IGDIMN demonstrates excellent classification performance in most of the initial classes, which is far higher than that of LwF and End2End. iCaRL performs close to IGDIMN in some incremental classes, but the memory ability for the initial classes is poor. FOSTER achieves the accuracy of 87.47% in the Hippo grass class, but the performance fluctuates greatly in certain classes. The comprehensive performance is similar to that of iCaRL, while both are slightly outperformed by IGDIMN. DCPRN performs outstandingly in some of the initial classes, but the adaptability to incremental classes is limited. PASS performs best in the Reeds class with the accuracy of 94.63%, yet the overall performance is unsatisfactory, especially in some initial classes, with the Kappa as high as 35.40%, indicating the insufficient adaptability to complex scenarios. Through comparative analysis of OA and AFR, PASS shows an AFR of 53.32% and an OA of 65.68%, indicating severe forgetting of old knowledge and

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TABLE IV
COMPARISON OF CLASS PERFORMANCE OF DIFFERENT APPROACHES FOR THE HOUSTON DATA

| | Class | LwF | iCaRL | End2End | PASS | FOSTER | DCPRN | IGDIMN |
|---|---|---|---|---|---|---|---|---|
| Initial Classes | Grass-stressed | 35.36 ± 11.24 | 61.50 ± 10.53 | 42.46 ± 5.26 | 69.67 ± 22.04 | 52.91 ± 14.85 | 25.81 ± 16.47 | **89.09 ± 7.24** |
| | Tree | 39.73 ± 18.94 | 70.51 ± 6.68 | 60.16 ± 7.12 | 66.57 ± 18.07 | 64.25 ± 19.59 | **90.07 ± 1.84** | 87.87 ± 7.96 |
| | Water | 8.70 ± 17.57 | 48.74 ± 21.11 | 8.63 ± 14.55 | 41.03 ± 15.47 | 72.26 ± 4.08 | 93.90 ± 3.77 | **95.14 ± 3.81** |
| | Commercial | 27.86 ± 5.72 | 51.54 ± 14.42 | 32.89 ± 1.40 | 9.73 ± 10.91 | 31.57 ± 3.37 | 39.18 ± 19.42 | **66.00 ± 4.51** |
| | Railway | 43.20 ± 9.92 | 20.62 ± 6.71 | 47.12 ± 8.14 | 50.68 ± 22.00 | 41.98 ± 12.95 | 3.56 ± 3.15 | **77.48 ± 4.34** |
| | Parking lot 2 | 18.86 ± 6.59 | 65.88 ± 17.08 | 33.08 ± 10.85 | 0.38 ± 0.62 | 71.94 ± 9.98 | 55.64 ± 3.74 | **94.37 ± 3.76** |
| | Tentis court | 99.17 ± 0.87 | 72.10 ± 7.64 | **99.43 ± 1.14** | 4.47 ± 9.99 | 96.21 ± 3.51 | 25.30 ± 17.95 | 86.54 ± 11.73 |
| | Running track | 4.44 ± 6.08 | 56.55 ± 13.27 | 29.06 ± 15.10 | 89.66 ± 11.52 | 87.07 ± 12.04 | 98.79 ± 0.87 | **99.12 ± 0.70** |
| Incremental Classes | Grass-healthy | 7.69 ± 3.05 | **93.72 ± 7.51** | 6.80 ± 2.70 | 91.19 ± 2.85 | 86.55 ± 3.84 | 99.52 ± 0.87 | 93.27 ± 7.40 |
| | Grass-synergic | 19.62 ± 15.00 | 99.80 ± 0.13 | 4.98 ± 3.92 | **99.90 ± 0.21** | 99.36 ± 0.60 | 99.78 ± 0.09 | 86.77 ± 12.83 |
| | Soil | 16.82 ± 17.99 | 98.89 ± 0.67 | 3.29 ± 7.06 | 93.67 ± 3.98 | **99.91 ± 0.00** | 99.77 ± 0.14 | 97.60 ± 3.25 |
| | Residential | 12.15 ± 4.65 | 85.65 ± 4.20 | 2.82 ± 4.22 | 60.86 ± 9.01 | 85.08 ± 9.86 | **94.93 ± 2.03** | 86.54 ± 7.11 |
| | Road | 4.33 ± 3.52 | 93.56 ± 3.14 | 6.05 ± 4.35 | 59.47 ± 12.07 | 67.97 ± 5.30 | **93.93 ± 1.30** | 91.53 ± 2.05 |
| | Highway | 8.21 ± 15.35 | 84.65 ± 14.58 | 0.54 ± 0.76 | 74.31 ± 20.55 | 85.45 ± 11.53 | **99.37 ± 0.51** | 96.32 ± 2.26 |
| | Parking lot 1 | 6.41 ± 4.19 | 96.37 ± 0.77 | 0.47 ± 0.47 | 86.79 ± 10.03 | 95.96 ± 3.45 | **99.66 ± 0.08** | 96.95 ± 2.06 |
| Performance | OA-INIT | 88.30 ± 1.06 | 69.87 ± 6.33 | 87.66 ± 1.23 | **99.71 ± 0.24** | 75.22 ± 5.37 | 99.85 ± 0.12 | 99.54 ± 0.20 |
| | OA-INIL | 34.84 ± 6.90 | 53.93 ± 2.42 | 44.81 ± 2.12 | 46.58 ± 3.00 | 57.40 ± 3.06 | 48.22 ± 0.91 | **84.03 ± 1.11** |
| | OA | 21.43 ± 1.00 | 75.09 ± 1.29 | 22.35 ± 1.00 | 64.50 ± 3.13 | 74.00 ± 1.91 | 75.33 ± 0.42 | **88.95 ± 1.42** |
| | AFR | 53.46 ↓ | 15.94 ↓ | 42.85 ↓ | 53.13 ↓ | 17.82 ↓ | 51.63 ↓ | **15.51 ↓** |
| | Kappa-ITP | 86.13 ± 1.25 | 64.87 ± 7.19 | 85.36 ± 1.46 | **99.66 ± 0.28** | 70.88 ± 6.32 | 99.83 ± 0.14 | 99.51 ± 0.21 |
| | Kappa-ILP | 15.33 ± 2.81 | 73.07 ± 1.42 | 16.44 ± 0.90 | 61.50 ± 3.41 | 71.93 ± 2.05 | 73.31 ± 0.46 | **88.47 ± 1.54** |

relatively limited performance in new class recognition. Both LwF and End2End demonstrate limited performance in terms of both metrics. In comparison, iCaRL, DCPRN, and FOSTER achieve lower AFR values of 5.97%, 6.31%, and 10.46%, with OA values of 73.84%, 90.48%, and 73.43%, respectively. These methods outperform PASS in overall classification performance, yet remain inferior to IGDIMN. In contrast, IGDIMN achieves the OA of 94.27% with the AFR of only 7.17%, demonstrating effective learning of new classes while significantly alleviating knowledge forgetting, which reflects stronger incremental learning capability.

In the last experiment on the Houston dataset, eight classes are selected in the initial stage, including Grass-stressed, Tree, Water, Commercial, Railway, Parking lot 2, Tennis court, and Running track. As shown in Table IV and Fig. 10, IGDIMN outperforms other methods both in overall classification performance and the maintenance of memory for initial classes. The LwF and End2End yield results that are less satisfactory compared to other models. For example, End2End only achieves the accuracy of 4.98% in the Grass-synergic class. Among other methods, the comprehensive performance of iCaRL, PASS, and FOSTER is significantly lower than that of DCPRN. Although DCPRN performs excellently in some incremental classes, the problem of forgetting the initial classes is quite obvious. In terms of comprehensive performance, the OA of IGDIMN and the Kappa coefficient lead comprehensively. The OA is higher than that of DCPRN, which is the second-best model, and the Kappa coefficient shows some improvement. Values of the AFR show that LwF, PASS, and DCPRN all exceed 50%, indicating substantial forgetting of old class knowledge during incremental learning. End2End also demonstrates considerable knowledge forgetting, with an AFR of 42.85%. By comparison, iCaRL, FOSTER, and the proposed IGDIMN achieve clearly lower AFR levels, which reflects the effective ability to mitigate catastrophic forgetting. In particular, IGDIMN attains the lowest AFR of 15.51%, demonstrating well retention of old-class knowledge compared to all methods. In conclusion, owing to the advantages of high stability and comprehensive performance, IGDIMN significantly outperforms other comparative models, verifying the effectiveness of IGDIMN in balancing new and old classes in incremental learning for HSI classification.

Additionally, Fig. 11 illustrates the grouped accuracy comparison based on four representative OA evaluation criteria, including OA-INIT, OA-INIL, OA-ICIL, and OA-ILSM.

On the Salinas dataset, PASS, DCPRN, and End2End all exceed 96% in OA-INIT, demonstrating good initial recognition of old land-cover classes. In the OA-INIL metric, IGDIMN achieves 95.51%, effectively retaining old hyperspectral land-cover classes. FOSTER and iCaRL are around 77%, while LwF and PASS perform unsatisfactorily. For new class recognition, DCPRN and PASS exceed 93% in OA-ICIL, showing good adaptability in incremental learning of new hyperspectral land-cover classes. IGDIMN achieves 85.96%, remaining at the relatively high level, while End2End and LwF exhibit limited performance. IGDIMN reaches 89.55% in OA-ILSM, maintaining stable incremental classification across different land-cover types in the Salinas scene.

On the Houston dataset, IGDIMN achieves 99.54% in the OA-INIT metric, showing strong performance in recognizing old hyperspectral land-cover classes in the ITP. DCPRN and PASS also reach values close to 100%, indicating similarly high effectiveness in the recognition of old class. In contrast, the accuracy of LwF and iCaRL is unsatisfactory, with End2End performing slightly better than the two methods.

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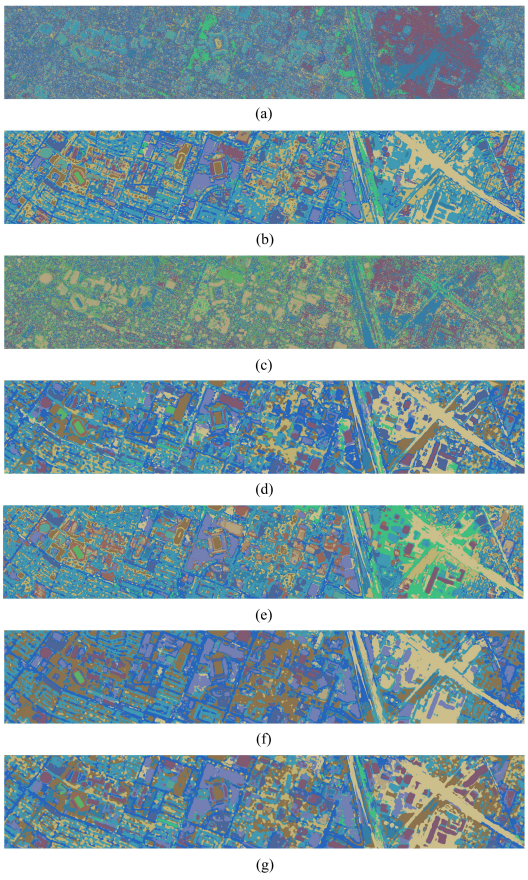



Fig. 10. Classification maps obtained via different HSI incremental classification methods for the Houston data. (a) LwF. (b) iCaRL. (c) End2End. (d) PASS. (e) FOSTER. (f) DCPRN. (g) IGDIMN.

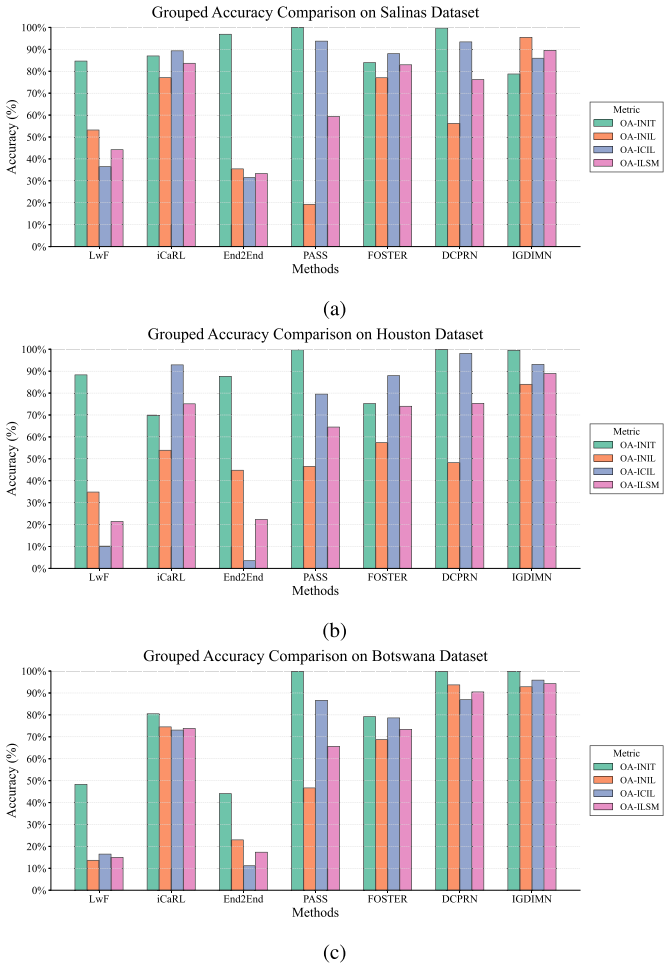



Fig. 11. Grouped accuracy comparison of different methods on three datasets based on four OA evaluation metrics: OA-INIL, OA-INIT, OA-ICIL, and OA-ILSM. (a) Salinas. (b) Houston. (c) Botswana.

In terms of OA-INIL, IGDIMN maintains the accuracy of 84.03%, indicating good retention of known classes in the ILP. FOSTER and DCPRN also exhibit good old class retention, while LwF and End2End fall below 45%, showing unsatisfactory performance. Regarding new class recognition, DCPRN achieves 98.10% in OA-ICIL and IGDIMN reaches 93.08%, both performing well in recognizing new hyperspectral categories in the ITP. LwF and End2End show limited effectiveness in recognizing new classes. IGDIMN achieves 88.95% in OA-ILSM, demonstrating the balanced performance between old class retention and new class recognition, and meeting the demands of incremental classification for complex land-cover types in HSI.

On the Botswana dataset, IGDIMN, DCPRN, and PASS all achieve 100% in OA-INIT, showing excellent accuracy in recognizing old hyperspectral classes in the ITP. FOSTER and iCaRL are near 80%, while LwF and End2End demonstrate much less effective performance. In OA-INIL, IGDIMN reaches 92.83% and DCPRN reaches 93.69%, reflecting strong old class retention and the ability to recognize learned hyperspectral classes as new categories are introduced. FOSTER and iCaRL perform slightly less effectively, while LwF and End2End remain unsatisfactory. For OA-ICIL, IGDIMN achieves 95.87% and DCPRN 86.94%, showing good performance in recognizing new hyperspectral classes, while FOSTER performs slightly worse. The remaining methods generally perform poorly in the metric. IGDIMN achieves 94.27% in OA-ILSM, demonstrating stable performance in incremental classification with continuously recognized hyperspectral categories.

### D. Ablation Study

Ablation studies are conducted on three HSI datasets to evaluate the influence of the proposed modules in the IGDIMN model. The configuration without any module serves as a baseline. Table V presents performance across two evaluation metrics, including OA-ICIL and OA-ILSM. Results confirm that each module contributes positively to model performance. On the Salinas dataset, integration of IGD improves new-class accuracy from 81.90% to 84.85% and overall-category accuracy from 85.35% to 86.80%. PAMTM module brings measurable gains, reaching 84.60% and 86.26% on the two metrics. Deployment of both modules together leads to further improvement, with overall-category accuracy increasing to 89.10%, indicating clear synergy between the two components. On the Houston dataset, IGD produces significant gains, elevating new-class accuracy to 96.22% and OA to 89.71%. Although individual application of PAMTM results

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TABLE V
ABLATION STUDY ON THE THREE HSI DATASETS

| | Module | | | Accuracy | |
|---|---|---|---|---|---|
| | Baseline | IGD | PAMTM | OA-ICIL | OA-ILSM |
| Salinas | ✓ | | | 81.90 | 85.35 |
| | ✓ | ✓ | | 84.85 | 86.80 |
| | ✓ | | ✓ | 84.60 | 86.26 |
| | ✓ | ✓ | ✓ | 84.86 | 89.10 |
| Houston | ✓ | | | 90.70 | 83.15 |
| | ✓ | ✓ | | 96.22 | 89.71 |
| | ✓ | | ✓ | 91.95 | 85.51 |
| | ✓ | ✓ | ✓ | 96.10 | 91.37 |
| Botswana | ✓ | | | 80.44 | 72.91 |
| | ✓ | ✓ | | 91.45 | 82.48 |
| | ✓ | | ✓ | 83.35 | 73.80 |
| | ✓ | ✓ | ✓ | 94.49 | 94.70 |

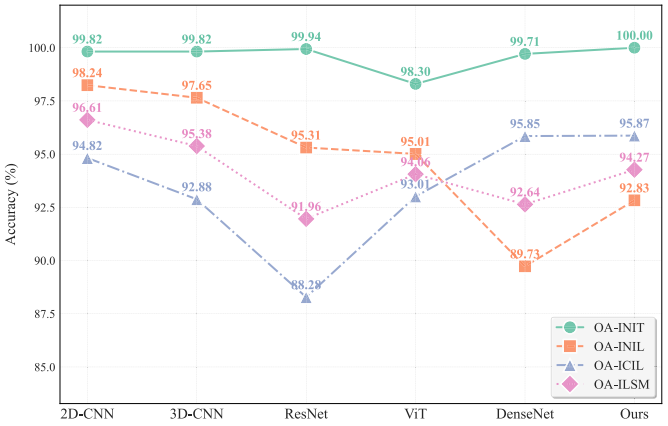



Fig. 12. Performance evaluation of the proposed IGDIMN framework with different backbone networks on the Botswana dataset.

in the modest score increase, combination with IGD yields 96.10% on novel classes and 91.37% on all classes, reflecting a complementary relationship. On the Botswana dataset, baseline performance on all categories remains at 72.91%. Incorporating IGD raises the value to 82.48%, and PAMTM brings additional improvement. When both modules are activated simultaneously, new-class accuracy rises to 94.49% and full-category accuracy reaches 94.70%, showing a substantial advantage in scenarios involving limited data.

To further validate network effectiveness, experiments on the Botswana dataset combined the core DGNet-DD framework of IGDIMN method with five backbone networks, including 2D-CNN, 3D-CNN, ResNet, Vision Transformer, and DenseNet. Results are shown in Fig. 12. As observed, the OA-INIT for all the compared backbones in the Botswana datasets is satisfactory, indicating that the basic initial backbones are effective for spectral–spatial feature extraction. Besides, the OA-ILSM of all the compared backbones is impressive, with the value of 96.61% and 95.38% for the 2D-CNN and 3D-CNN. Experimental results demonstrate that under the DGNet-DD framework, all compared backbone networks effectively mitigate forgetting of old classes while learning new classes.

Overall results show that IGD and PAMTM alleviate catastrophic forgetting while maintaining learning ability on novel categories. Integration of both modules facilitates effective fusion of previously acquired and newly learned knowledge, consistently achieving high performance across three datasets and demonstrating strong robustness in incremental learning.

### E. Analysis of the Dual-Graph Interaction

To evaluate the effectiveness of the single-graph representation mechanism and the dual-graph interaction mechanism within the IGDIMN framework for HSI incremental classification, we conducted a series of comparative experiments across three HSI datasets. The evaluation was based on four key metrics, including OA-INIT, OA-INIL, OA-ICIL, and OA-ILSM.

The performance of the single-graph representation mechanism and the dual-graph interaction mechanism is illustrated through visual comparisons for intuitive understanding. Fig. 13(a)–(c) corresponds to the results on the Salinas, Houston, and Botswana datasets, respectively. The dual-graph mechanism consistently outperforms the single-graph setting across all datasets. On the Salinas dataset, the dual-graph setting also showed superior performance, with 95.88% in OA-INIL and 89.78% in OA-ILSM, compared to 81.97% and 76.60% achieved by the single-graph model. On the Houston dataset, the dual-graph model achieved 79.53% in OA-INIL and 87.66% in OA-ILSM, while the single-graph model only reached 73.13% and 78.47%, respectively. On the Botswana dataset, the dual-graph model exceeded 94% across all four metrics, significantly outperforming the single-graph model with OA-INIL and OA-ILSM values of 74.65% and 80.05%. The experimental results demonstrate that the dual-graph interaction mechanism provides strong support for preserving old class knowledge and adapting to new classes, thereby improving the overall performance and robustness of the HSI incremental classification.

### F. Analysis of Experimental Parameters

In this section, the experiment evaluates the impact of different experimental parameters on the classification accuracy of IGDIMN. First, we analyzed the impact of different patch sizes on IGDIMN. Each dataset is tested with four distinct patch sizes: $9\times9$, $11\times11$, $13\times13$, and $15\times15$. The evaluation metrics include OA-INIL, OA-INIT, OA-ICIL, and OA-ILSM. Fig. 14(a)–(c) corresponds to the results on the three datasets.

On the Salinas dataset, classification performance increases progressively with larger patch sizes. With a patch size of $9\times9$, OA-INIL reaches 95.88% and OA-ILSM reaches 89.78%, indicating moderate accuracy. As the patch size increases, the values for $15\times15$ reach 96.75% for OA-INIL and 89.99% for OA-ILSM, showing the clear improvement over $11\times11$ and $13\times13$. On the Houston dataset, the patch size of $15\times15$ provides the highest accuracy, with OA-INIL at 88.57% and OA-ILSM at 91.79%, outperforming the smaller sizes. In contrast, the $9\times9$ setting results in lower scores, with OA-INIL at 82.55% and OA-ILSM at 89.25%. For the Botswana dataset, the $15\times15$ patch size also leads to the best results, reaching 97.42% in OA-INIL and 93.23% in OA-ILSM. Comparatively,

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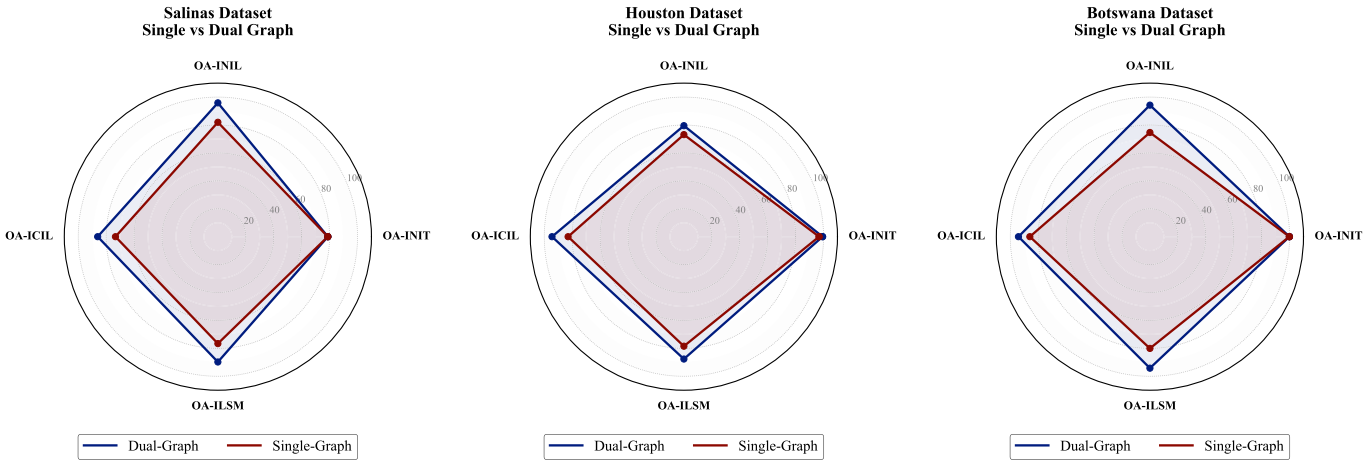



Fig. 13. Accuracy comparison of single graph and dual graph on three datasets across four OA metrics. (a) Salinas. (b) Houston. (c) Botswana.

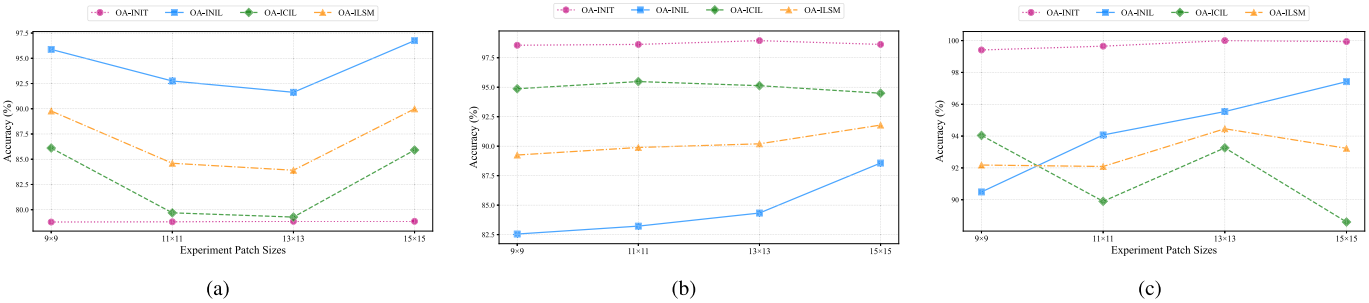



Fig. 14. Accuracy comparison of different patch sizes on three datasets across four OA metrics. (a) Salinas. (b) Houston. (c) Botswana.

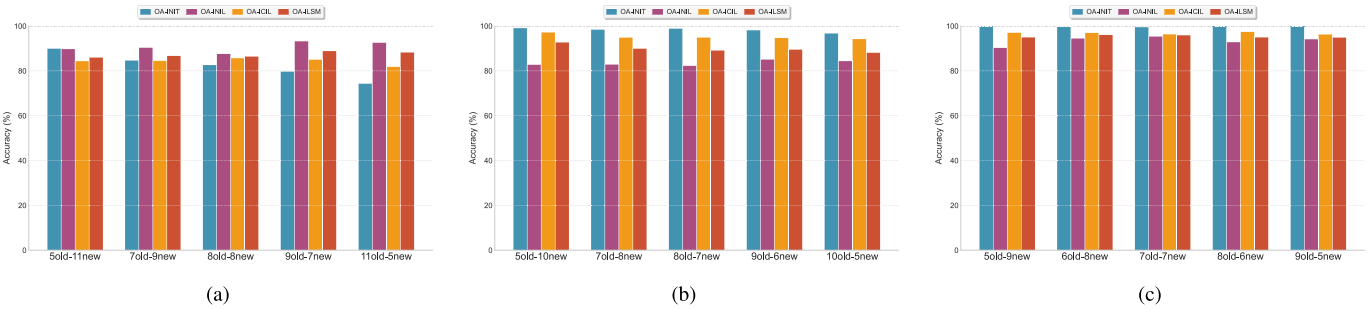



Fig. 15. Accuracy comparison under different class configurations on three datasets. (a) Salinas. (b) Houston. (c) Botswana.

the $9 \times 9$ configuration shows reduced accuracy, with OA-INIL at 90.49% and OA-ILSM at 92.18%.In summary, larger patch sizes substantially improve the classification performance of IGDIMN, particularly in terms of OA-INIL and OA-ILSM. As the patch size increases, IGDIMN becomes effective at capturing spatial–spectral features, thereby enhancing classification accuracy. Although $9 \times 9$ and $11 \times 11$ yield reasonable performance on some datasets, the $15 \times 15$ setting provides the most consistent and superior results overall.

Second, we investigated the impact of different numbers of old and new classes on IGDIMN. Fig. 15(a)–(c) corresponds to the results on the three datasets. The Salinas dataset was tested under five configurations, including 5old-11new, 7old-9new, 8old-8new, 9old-7new, and 11old-5new. Experimental results show that the model achieves optimal overall performance when old classes outnumber new classes in the 9old-7new configuration. As can be seen, our approach model excelled in the 9old-7new configuration with OA-INIL of 93.32%, highlighting superior retention of old classes. On the Houston dataset, five class configurations were evaluated, including 8old-7new, 10old-5new, 9old-6new, 7old-8new, and 5old-10new. As observed, our model achieved the best performance with the setting of 5old-10new with OA-INIT of 99.16%, which demonstrates the effective initial learning of our approach. With an OA-ICIL of 97.29%, the proposed model shows a robust ability to learn new classes. The Botswana dataset experiment encompassed five configurations, including 6old-8new, 8old-6new, 5old-9new, 9old-5new, and 7old-7new. The model performed best with the setting of 6old-8new with OA-

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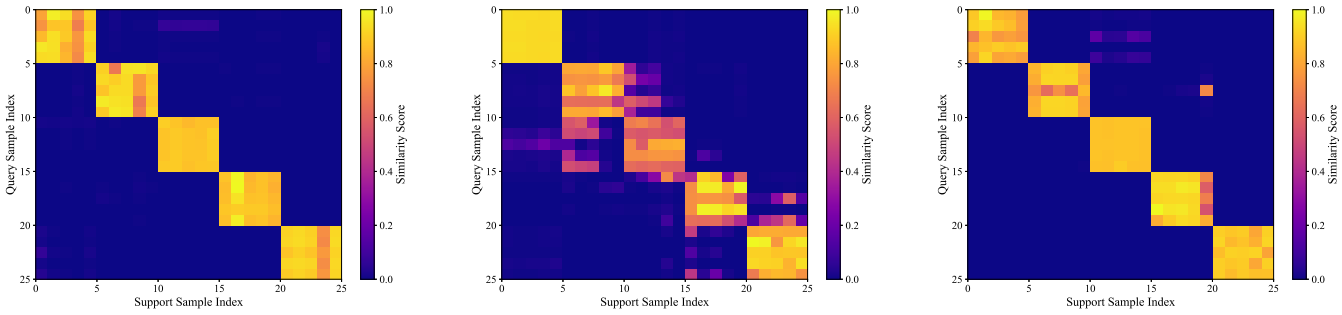



Fig. 16. Similarity heatmaps between query and support sets on three datasets. (a) Salinas. (b) Houston. (c) Botswana.

TABLE VI
COMPARISON OF IGDIMN-ITP AND IGDIMN-ILP PERFORMANCE ACROSS THREE EXPERIMENTS

| | Experiment 1 | | | Experiment 2 | | | Experiment 3 | | |
|---|---|---|---|---|---|---|---|---|---|
| | Class | Baseline-ITP | Baseline-ILP | Class | Baseline-ITP | Baseline-ILP | Class | IGDIMN-ITP | IGDIMN-ILP |
| Initial Classes | Asphalt_C1 | 93.54 ± 0.15 | 57.59 ± 2.39 | Asphalt_C1 | 95.07 ± 0.17 | 83.45 ± 0.99 | Asphalt_C1 | 94.86 ± 0.47 | 75.62 ± 1.93 |
| | Meadows_C1 | 99.26 ± 0.26 | 92.66 ± 0.67 | Meadows_C1 | 99.39 ± 0.47 | 84.85 ± 0.54 | Meadows_C1 | 99.46 ± 0.12 | 94.92 ± 1.20 |
| | Trees_C1 | 97.00 ± 0.10 | 83.55 ± 0.45 | Trees_C1 | 96.55 ± 0.26 | 84.61 ± 0.56 | Trees_C1 | 96.45 ± 0.30 | 86.99 ± 5.52 |
| | Bare Soil_C1 | 98.21 ± 0.55 | 92.51 ± 0.99 | Bare Soil_C1 | 97.89 ± 0.75 | 84.36 ± 0.80 | Bare Soil_C1 | 97.79 ± 0.94 | 95.72 ± 1.56 |
| | Bitumen_C1 | 97.37 ± 0.26 | 78.82 ± 3.41 | Bitumen_C1 | 97.42 ± 0.26 | 84.49 ± 0.73 | Bitumen_C1 | 97.40 ± 0.41 | 95.93 ± 1.62 |
| | Self-Blocking Bricks_C1 | 99.81 ± 0.13 | 62.08 ± 0.48 | Self-Blocking Bricks_C1 | 99.73 ± 0.21 | 80.23 ± 1.26 | Self-Blocking Bricks_C1 | 99.52 ± 0.39 | 91.49 ± 4.94 |
| | Shadows_C1 | 99.06 ± 0.64 | 65.24 ± 1.24 | Shadows_C1 | 99.25 ± 0.57 | 83.53 ± 0.57 | Shadows_C1 | 99.56 ± 0.22 | 93.28 ± 4.95 |
| Incremental Classes | Asphalt_U | – | 88.90 ± 1.18 | Asphalt_C2 | – | 84.67 ± 0.95 | Asphalt_U | – | 95.24 ± 2.06 |
| | Meadows_U | – | 71.70 ± 1.71 | Meadows_C2 | – | 86.45 ± 0.69 | Meadows_U | – | 92.76 ± 3.14 |
| | Trees_U | – | 93.58 ± 0.34 | Trees_C2 | – | 83.69 ± 0.90 | Trees_U | – | 95.76 ± 1.50 |
| | Bare Soil_U | – | 83.98 ± 1.99 | Bare Soil_C2 | – | 82.65 ± 1.00 | Bare Soil_U | – | 99.38 ± 0.34 |
| | Bitumen_U | – | 78.17 ± 2.14 | Bitumen_C2 | – | 84.09 ± 0.83 | Bitumen_U | – | 98.66 ± 0.30 |
| | Self-Blocking Bricks_U | – | 93.05 ± 1.35 | Self-Blocking Bricks_C2 | – | 82.70 ± 0.67 | Self-Blocking Bricks_U | – | 97.80 ± 1.71 |
| | Shadows_U | – | 87.37 ± 2.06 | Shadows_C2 | – | 82.95 ± 0.86 | Shadows_U | – | 97.63 ± 0.86 |
| Performance | OA-INIT | 96.98 ± 0.20 | – | OA-INIT | 97.22 ± 0.17 | – | OA-INIT | 97.15 ± 0.42 | – |
| | OA | – | 78.23 ± 0.51 | OA | – | 83.91 ± 0.24 | OA | – | 91.93 ± 0.84 |
| | OA-INIL | – | 75.99 ± 0.30 | OA-INIL | – | 83.92 ± 0.24 | OA-INIL | – | 88.82 ± 1.64 |
| | OA-ICIL | – | 80.47 ± 0.77 | OA-ICIL | – | 83.92 ± 0.26 | OA-ICIL | – | 95.05 ± 1.22 |
| | AFR | – | 20.99 ↓ | AFR | – | 13.3 ↓ | AFR | – | 8.33 ↓ |
| | Kappa | 98.92 ± 0.13 | 99.22 ± 0.21 | Kappa | 82.42 ± 0.26 | 90.93 ± 0.95 | Kappa | 99.13 ± 0.51 | 90.93 ± 0.95 |

ILSM of 96.09% and OA-INIL of 94.6%. Notably, all the values of OA-INIT are close to 100% and show the robustness of our model in the initial learning phase on Botswana. Through the experiment across three datasets, we have verified the robustness and scalability of our model with various class configurations.

### G. Analysis of Support–Query Similarity Visualization

To analyze the evolution of classification performance of IGDIMN in HSI incremental classification within the meta-learning framework, similarity heatmaps between the support and query sets are generated to visualize the structural perception of the model in the incremental phase. The similarity results for the support and query sets from three datasets, Salinas, Houston, and Botswana, are shown in Fig. 16. Initially, the similarity between all query samples and the support set is set to the uniform value of $1/N$ (where $N$ is the number of support samples), without class-specific structural differences. The uniform similarity corresponds to the initial state of IGDIMN, where no class-discriminative ability has been learned. In Fig. 16(a)–(c), heatmaps of IGDIMN after training on different datasets are presented. A noticeable enhancement in similarity between query samples and same-class support samples can be observed, forming distinct patterns of high intraclass similarity and low interclass similarity. As training progresses, IGDIMN gradually learns the semantic relations among support samples and effectively transfers knowledge to the similarity estimation

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of query samples, thereby enhancing the ability to recognize class structures.

Furthermore, the results validate that the IGDIMN framework exhibits excellent generalization and incremental adaptability. The structural evolution of the heatmaps clearly demonstrates the dynamic balance between maintaining old-class recognition and adapting to new-class identification.

### H. Analysis of the Impact of Spectral Variability on IGDIMN

To analyze the effectiveness of spectral variability on IGDIMN, we conducted experiments on two HSI datasets, including Pavia Center (PC) and Pavia University (PU) by selecting seven same categories. In the subsequent three experiments, we randomly split the sample set of PC into two sets, denoted as C1 and C2. Notably, both sets contain the same number of samples and the same categories. We formed a new class U by selecting an equal number of samples from each category shared with the PC dataset within the PU dataset. The baseline network refers to our model without the IGD and PAMTM modules.

The first experiment employed C1 as the initial training set and U as the incremental set. In this situation, the two datasets exhibit an obvious spectral difference. The results are shown in Table VI. As observed, the initial OA-INIT is 96.98%, while the OA-INIL drops to 75.99% in the incremental phase with an AFR of 20.99%, which indicates the substantial degradation of knowledge regarding the old classes. The results verify the effect of the spectral variability on the incremental HSI classification. For the second experiment, C1 served as the initial set and C2 as the incremental set. As observed, the OA in the incremental phase is 83.91%, with an initial OA-INIT of 97.22% and an OA-INIL of 83.92%, resulting in an AFR of 13.3%. The accuracy for the old classes was generally high, and the decrease in AFR was much smaller than AFR in Table VI. The results show that catastrophic forgetting is less severe in the same distribution of the datasets in incremental classification of HSI. Furthermore, we conduct the third experiment using the same datasets and settings as the first experiment, and we employ our proposed model instead of the baseline model to implement the incremental classification. In this experiment, IGDIMN-ITP and IGDIMN-ILP mean the initial and incremental phases of the proposed IGDIMN method. The results presented in Table VI demonstrate that when the IGD and PAMTM strategies are employed during training, the OA in the incremental phase reaches 91.93% and the AFR decreased to 8.33%. These results verify that our model not only attains high initial accuracy but also demonstrates robust resistance to the catastrophic forgetting problem in incremental learning.

## IV. Conclusion

In this article, we propose IGDIMN to efficiently transfer old-class knowledge between the initial and ILPs in HSI incremental classification. SSDBN captures both spectral and spatial features of new and old classes through the dual-path multilayer convolution. Additionally, the core module DGNet-DD with IGD that adopts a dual-graph interaction mechanism consolidates old-class knowledge across training stages and mitigates forgetting in traditional methods in HSI incremental classification. Furthermore, without replaying old class samples, the proposed PAMTM aligns new class prototypes across incremental stages via the multilevel prototype strategy to reduce spectral distribution drift between old and new classes. Extensive experimental results and analyses demonstrate that the proposed model effectively transfers old-class knowledge in the incremental learning process, ensuring accurate recognition of new classes while retaining the memory of old-class knowledge in HSI incremental classification.

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